

Skills Summary

Area and Perimeter Without the Mix-Up

Use this reference while completing your worksheet or when studying.

Area: the Space Covered Inside

Rule: Area of a rectangle = length $\times$ width. Area is counted in *square* units: cm^2 , m^2 , square units.

Example: A rectangle is 8 cm long and 3 cm wide. What is its area?

Step 1. The question asks how much space the rectangle covers inside, so this is area: multiply length by width.

Step 2. $8 \times 3 = 24$.

Area: **24 cm^2**

Try it: A rectangle is 9 cm long and 4 cm wide. What is its area?

check: **36 cm^2**

Perimeter: the Distance Around

Rule: Perimeter of a rectangle = add *all four* sides: $P = 2 \times (\text{length} + \text{width})$. Perimeter is counted in plain length units: cm, m, ft.

Example: A rectangle is 7 cm long and 2 cm wide. What is its perimeter?

Step 1. The question asks for the distance around the edge, so add all four sides — two lengths and two widths.

Step 2. $2 \times (7 + 2) = 2 \times 9 = 18$.

Perimeter: **18 cm**

Try it: A rectangle is 5 m long and 4 m wide. What is its perimeter?

check: **18 m**

Which One Does the Story Need?

Rule: Ask: does the story *cover* something (paint, carpet, grass) — that is **area** — or *go around* something (fence, ribbon, frame) — that is **perimeter**?

Example: Paint covers a wall that is 5 m long and 3 m tall. Area or perimeter — and how much?

Step 1. Paint covers the whole face of the wall, so this story needs area, not the distance around.

Step 2. $5 \times 3 = 15$.

The paint covers **15 m^2**

Try it: A string of lights goes around a window that is 4 m long and 2 m tall. Area or perimeter — and how much?

check: perimeter — 12 m

(!) Watch out: Check the unit of your answer! Area answers need *square* units (cm^2 , m^2). Perimeter answers use plain length units (cm, m). A square unit on a fence — or a plain unit on a carpet — means the wrong tool was picked.